

CSCI 150 | EyeDHD

Introduction

The Psychology Department at Fresno State has been conducting a study utilizing a mixed reality simulation to find a way to diagnose ADHD using objective data, as opposed to subjective data. The tool we are creating would be using eye tracking data from a mixed reality simulation to assist in diagnosing ADHD and finding a possible bio-marker linked to ADHD. The purpose of this feasibility study is to assess both the technological requirements and the potential acceptance of this approach by combining research into the necessary tools and resources with feedback gathered from our student survey.

Tools and Resources

The project is broken down into four main components: front-end, back-end, storage, data, and other elements.

Front End	React
Back End	Node.js
Storage	PostgreSQL, Contabo VPS
Data	Matlab
Other	Git, Github

Survey Results

The survey results were supportive of our project, with average scores of (1-5): [3.5](#), [3.9](#), [3.5](#), [3.9](#), and [4.1](#) for questions one through five, respectively. The results indicated that most participants agreed this project would be useful. Opinions were more divided on whether the tool could aid in diagnosing ADHD, with responses split between neutral and agreement. Overall, the findings suggest that this tool could be highly valuable to the psychology community.

Alarmingly, one respondent **strongly disagreed** with Professor Herdman being the man — must have been a mistake. Opinions were more divided on whether the tool could aid in diagnosing ADHD, with responses split between neutral and agreement. Overall, the findings suggest that this tool could be highly valuable to the psychology community.

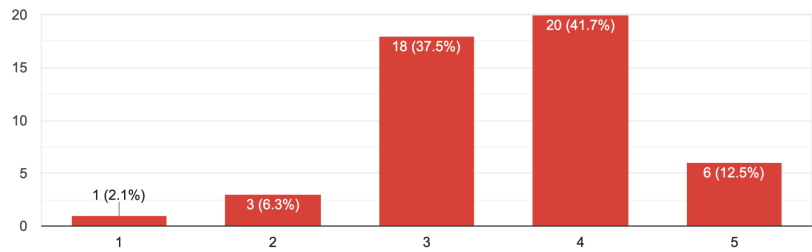
Conclusion

Using the given tools and resources, this project is certainly very feasible. The project may need adjustments for how the data is stored, but further communication with the Psychology department will determine this. These findings suggest that the EyeDHD project is both technically achievable and socially supported. The participants in the survey showed interest in technology being used for objective diagnoses in psychology, beyond ADHD. Overall, the feedback received from the survey suggests that the project has great value. The positive response indicates that this project can potentially become a practical tool in healthcare.

Question 1

I believe that using pupil dilation and eye movement data can provide objective and meaningful insights for diagnosing ADHD.

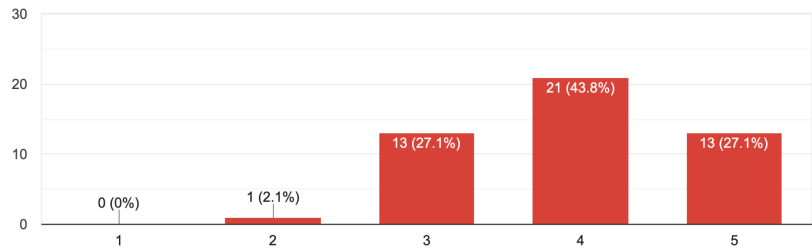
48 responses



Question 2

I believe this would be a useful tool for future ADHD research and related studies.

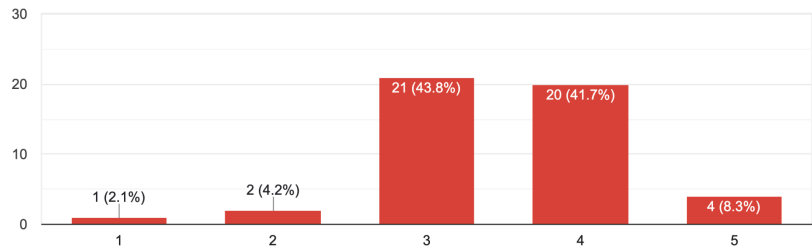
48 responses



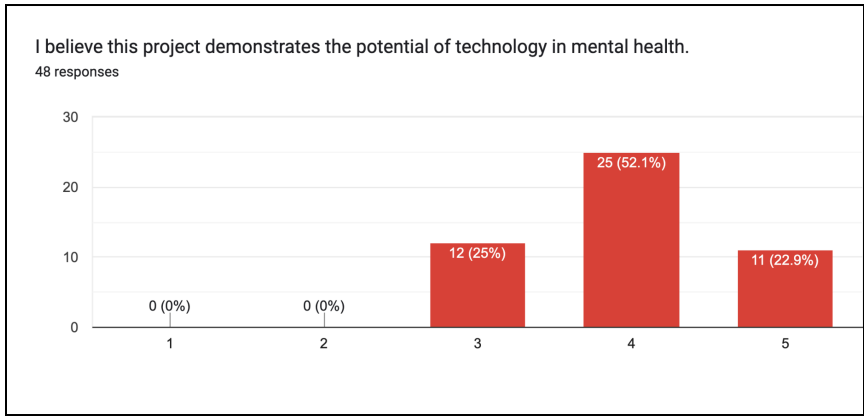
Question 3

I believe ADHD can be diagnosed using objective data rather than subjective data.

48 responses



Question 4



Question 5

